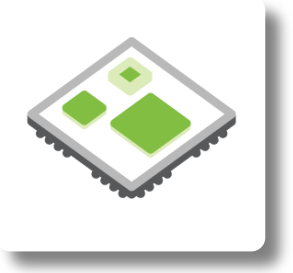
SERVER:
OPEN CHIPLET
ECONOMY



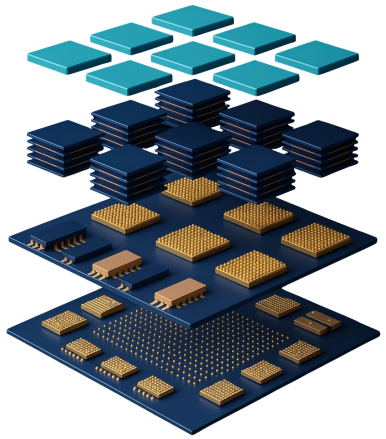
Building Interoperable AI Systems with Chiplets: Arm Neoverse and Cadence in an Open Ecosystem

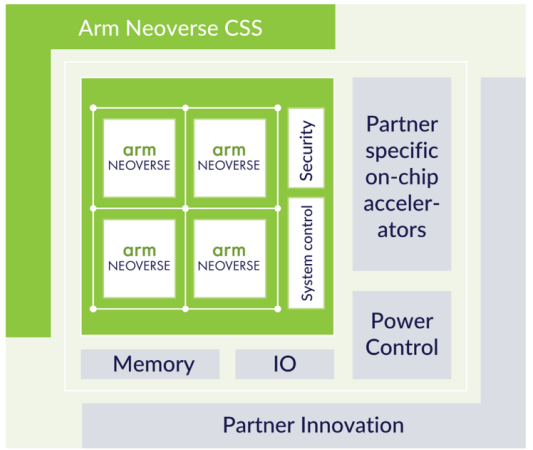
Saif Elsaady [Arm]

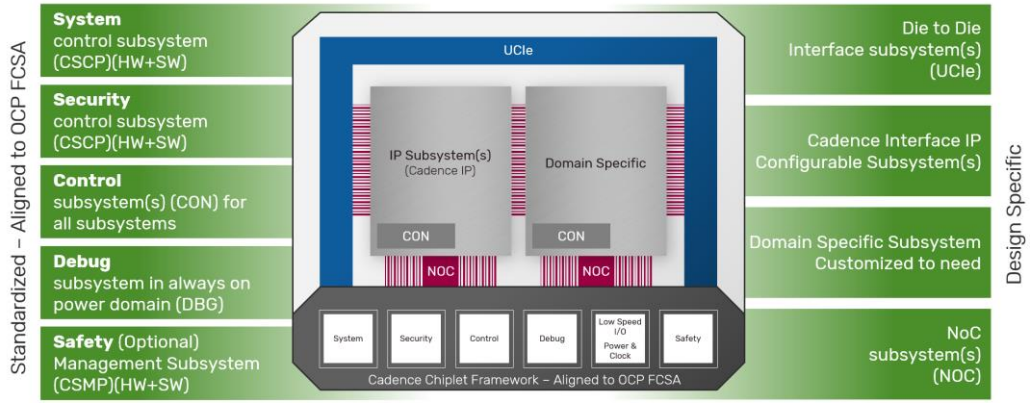
Michael Posner [Cadence Design Systems]

Agenda

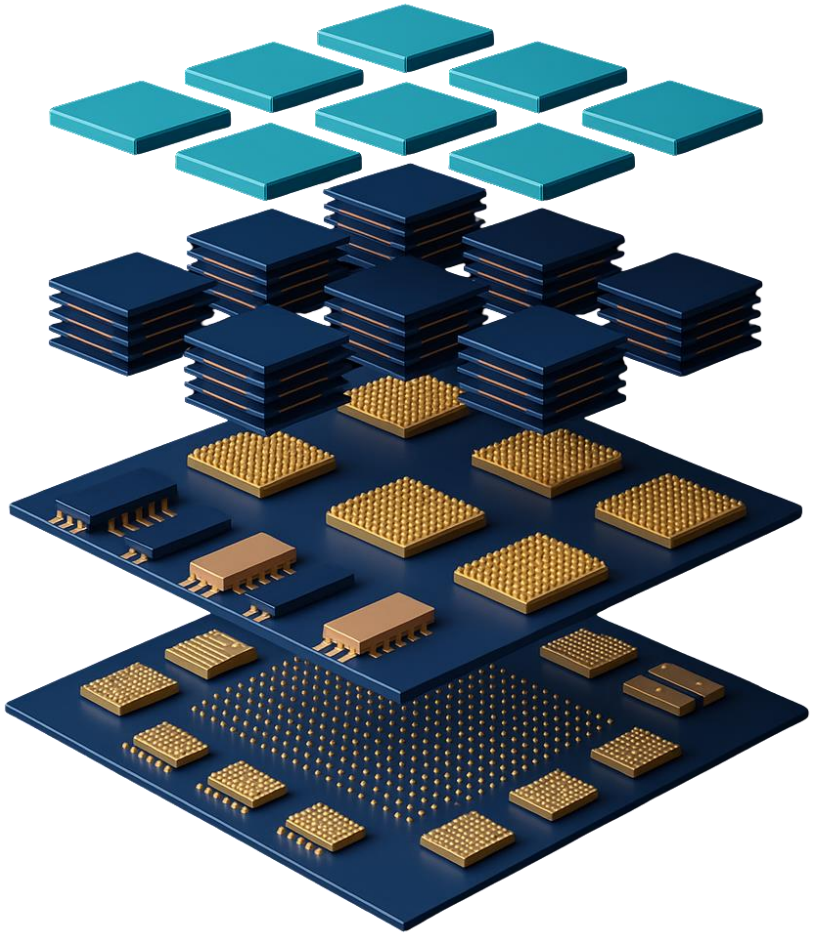
1. **Why chiplets for AI systems**
   Benefits of a Chiplet Approach
2. **Introducing the backbone of AI Systems**
   Arm Neoverse Compute Subsystems
3. **Shift Left Strategies**
   Pre-silicon Verification
4. **Enabling Interoperable chiplets**
   OCP Foundational Chiplet System Architecture
5. **Accelerating chiplet realization**
   Ready to utilize chiplet framework
6. **Conclusion**
   Summary and Call to Action







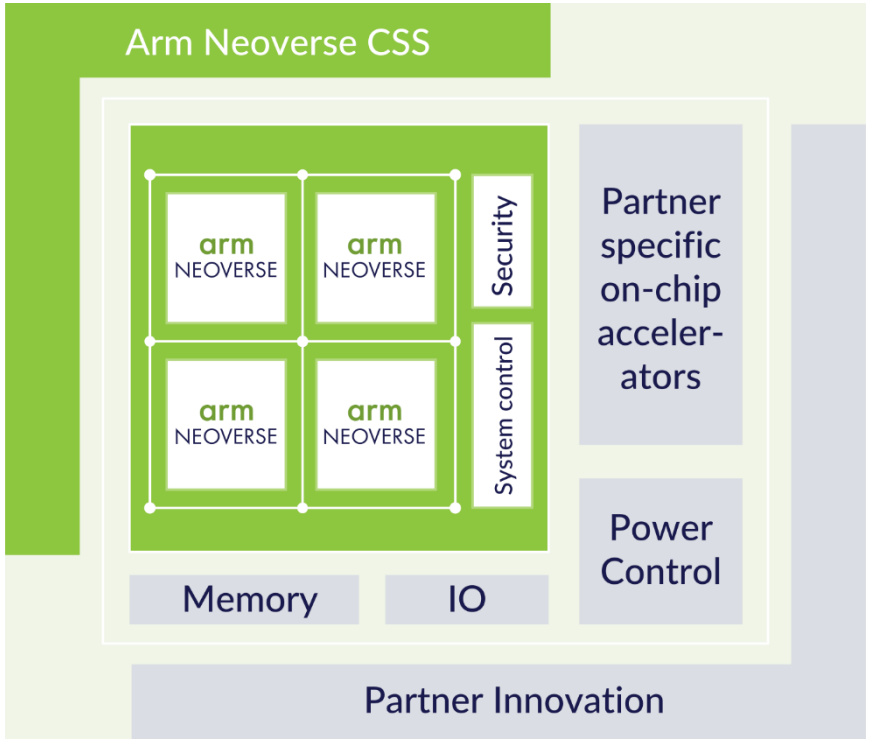
Why chiplets for AI systems?



- Leverage Investment
- Reduce Risk
- Tighter Integration

Arm Neoverse Compute Subsystems

A pre-integrated foundation for custom silicon



Arm provides

Neoverse CPUs, coherent interconnect and system IP

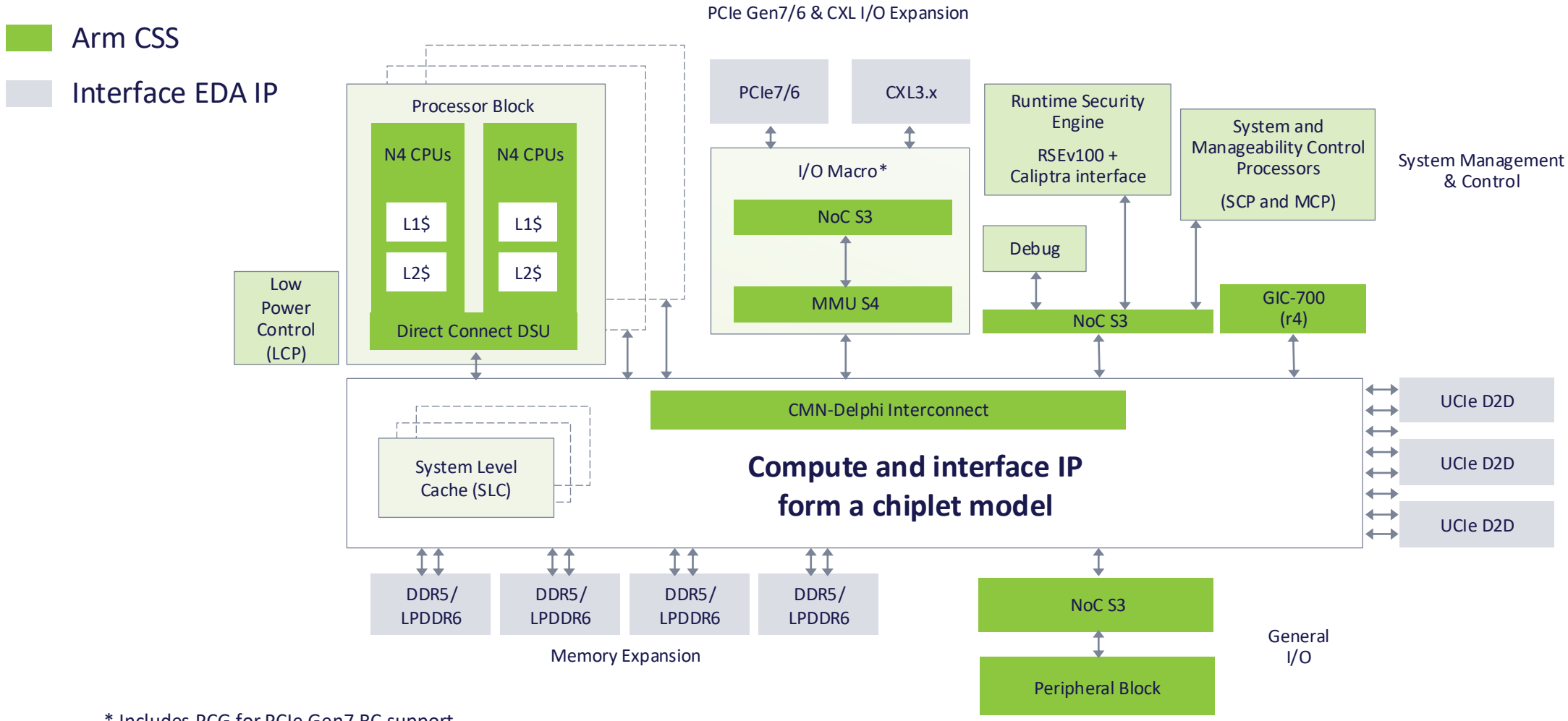
Verified integration and platform software

Partners add

I/O and accelerators, then complete the chip design

Less integration work. A faster path to custom silicon.

Arm Neoverse CSS N4 and interface IP integration

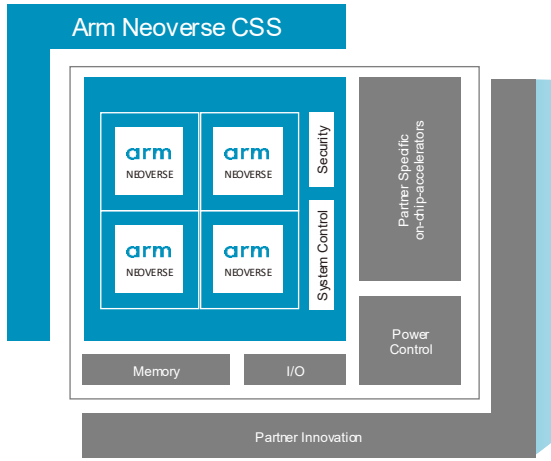


* Includes PCG for PCIe Gen7 RC support

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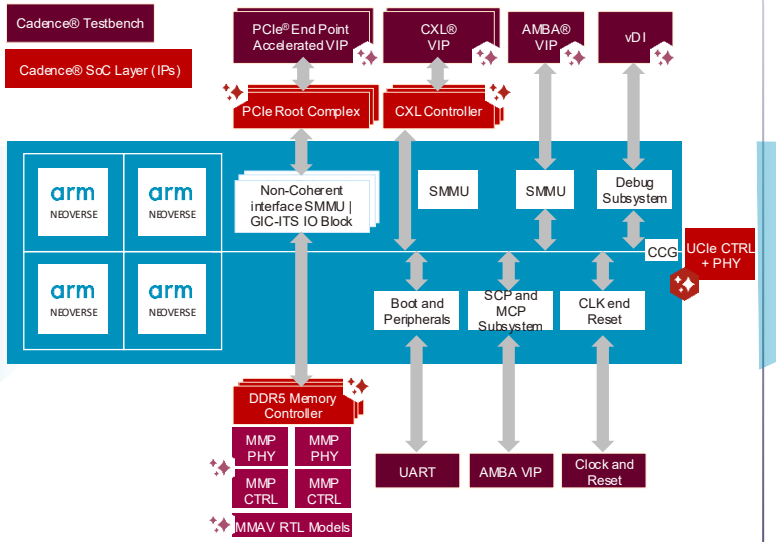
Shift Left with virtual prototyping, simulation and emulation

Building on Arm Foundation



Arm® Total Design Partnership

Enabled by Cadence IP & Tools

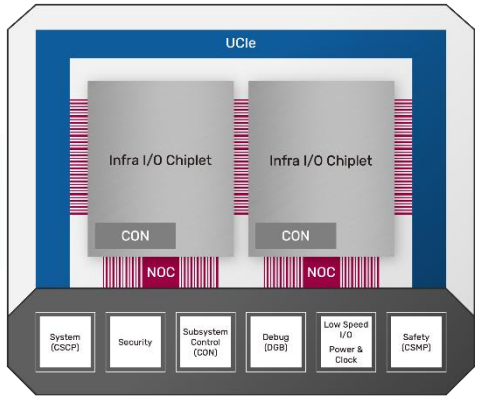


Helium™ Virtual Platform

Xcelium® Simulation

Palladium® Emulation

Accelerate Interoperable Chiplets



Accelerated Time To Market
Cadence Chiplet Framework and automation tools speed implementation

Customizable
Leverages Cadence chiplet reference platforms, IP, SoC and Chiplet expertise

Interoperable
Chiplets adhere to OCP FCSA

Early software development and system validation before silicon

Standardization Ensures Chiplets Work Together Seamlessly

Interoperability
Enables integration of chiplets from different vendors

Cost Efficiency
Reusable designs and processes

Faster Time to Market
Leverage standardized components

Ecosystem Growth
Jumpstart the chiplet marketplace

Arm and Cadence contributing to OCP Open Chiplet Economy



OCP Foundation Chiplet System Architecture (FCSA)

END USER

CHIPLETS VENDORS

MARKETPLACE

ASIC (SiP) BUILDERS

SUPPORT ECOSYSTEM

IP Provider | EDA | Manufacture | Tools

OPEN PLATFORMS
Standards, Tools, Business Workflows

Mature	Vendor-neutral
Architecture-agnostic	Common baseline for interoperable chiplets

Help drive standardization: Link to FCSA

Ready to Use Chiplet Manager

Cadence's Framework implementation of OCP FCSA

Standardized – Aligned to OCP FCSA

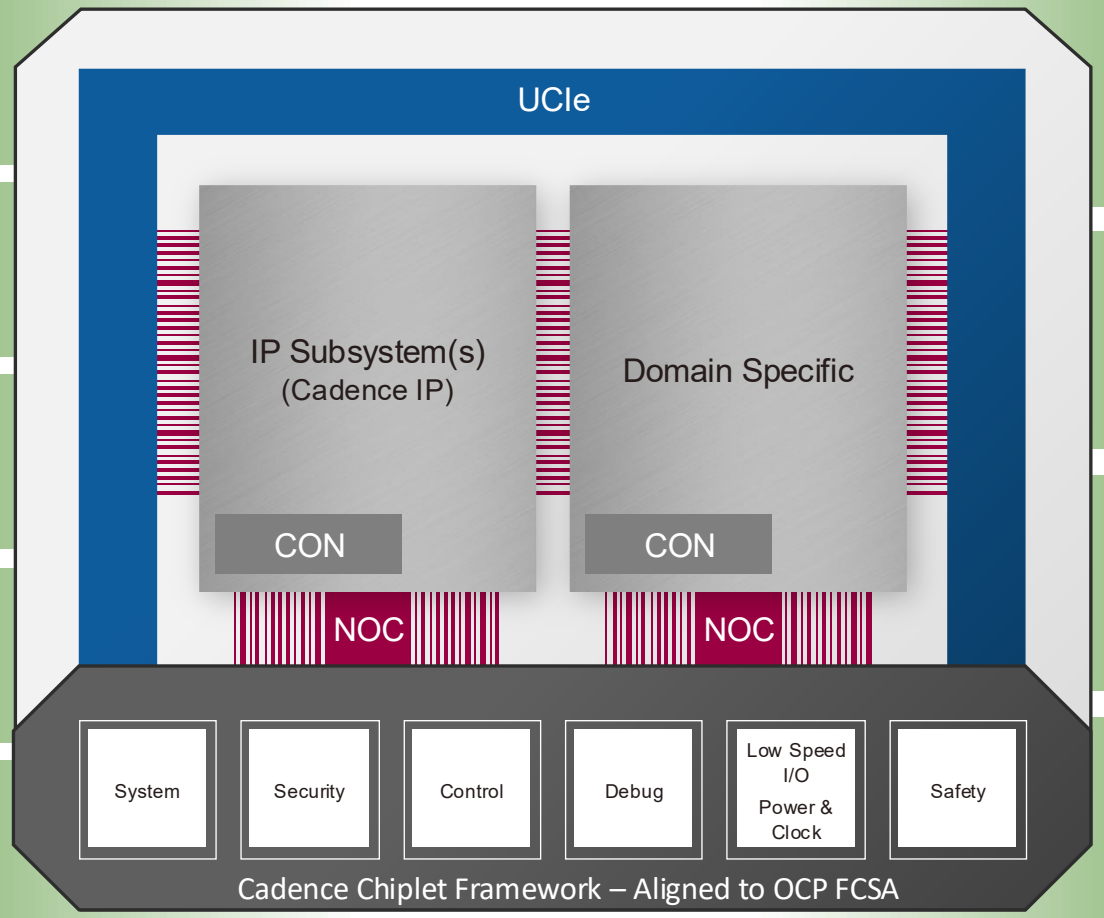
System control subsystem (CSCP)(HW+SW)

Security control subsystem (CSCP)(HW+SW)

Control subsystem(s) (CON) for all subsystems

Debug subsystem in always on power domain (DBG)

Safety (Optional) Management Subsystem (CSMP)(HW+SW)



Die to Die Interface subsystem(s) (UCIe)

Cadence Interface IP Configurable Subsystem(s)

Domain Specific Subsystem Customized to need

NoC subsystem(s) (NOC)

Design Specific

Cadence I/O Chiplets Example

I/O Interface Chiplet

UALink Ethernet Combo
Boot Configurable

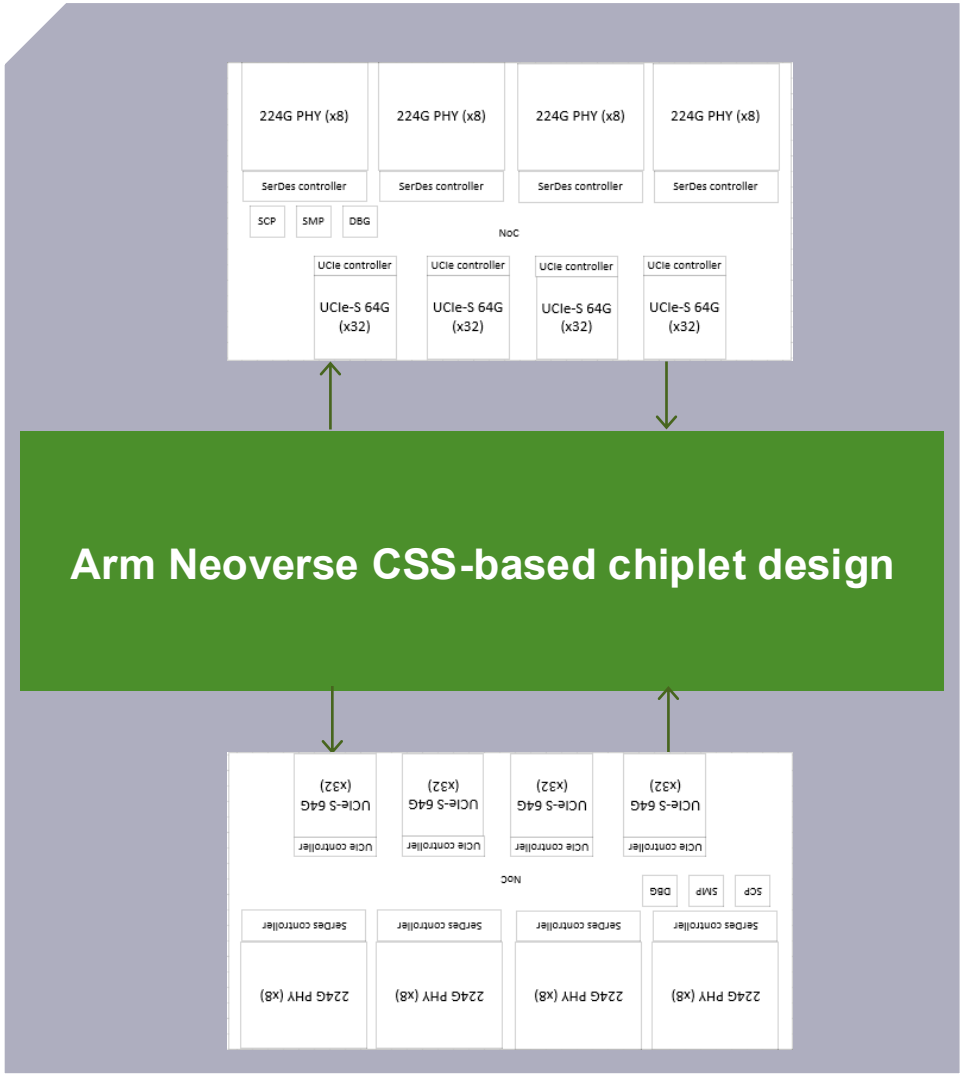
Cadence Chiplet
Framework for OCP FCSA aligned chiplet management capabilities

Each chiplet consists of :

4
Direct connect, low latency, slices of a UCIE/SerDes subsystem;

Each 224G SerDes x8 lanes for a total of x32 lanes

Supporting
UCIe-S or UCIe-A



Summary and Call to Action

Align compute, interfaces and early validation to build interoperable AI systems.

Ecosystem partners

Engage Arm and Cadence to align your compute, I/O and chiplet integration needs.

Implementation teams

Start software and system validation before silicon using the joint shift-left approach.

Standards community

Participate in OCP FCSA and help align chiplet interfaces, roles and services.

[Explore Arm Neoverse CSS]
[Explore Cadence chiplet solutions]

Thank You!

